



StudentHub

Requirements Analysis and System Design

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1 Introduction

1.1 Purpose

The purpose of StudentHub is to provide a platform designed by students, for students. Its objective is to streamline the management of student life. The platform will allow students to manage their tasks and organise their study sessions. The platform will also provide a space for students to connect with each other, share resources and collaborate on projects. The platform will also offer students a network forum where they can post announcements to find used books and private lessons, as well as share information about school events.

1.1.1 Goals

ID	Description
G1	Allow users to create, delete, log in to and log out of their accounts
G2	Enable users to create, edit, move, and delete deadlines on a shared calendar
G3	Provide functionality for users to create, edit, and delete private deadlines visible only to themselves
G4	Allow users to upload, share, edit, and delete study notes on various subjects
G5	Enable users to create, manage, and delete study groups with other platform users
G6	Provide a messaging system for communication within study groups
G7	Allow users to send and receive study group invitations
G8	Provide a forum where users can create, edit, and delete announcements
G9	Enable users to comment on and interact with forum announcements
G10	Allow users to search for specific topics, notes, or announcements within the platform
G11	Provide a marketplace functionality for exchanging used academic books
G12	Enable users to offer and search for private tutoring lessons
G13	Allow users to create and share announcements about upcoming school events
G14	Provide notification functionality for deadline reminders and forum activity
G15	Enable users to customize their profile and privacy settings

Table 1: Goals

1.2 Scope

1.2.1 World Phenomena

ID	Description
WP1	A student wants to create an account
WP2	A student wants to delete an account
WP3	A student wants to create a deadline on the shared/private calendar
WP4	A student wants to move a deadline on the shared/private calendar
WP5	A student wants to edit a deadline on the shared/private calendar
WP6	A student wants to delete a deadline on the shared/private calendar
WP7	A student wants to upload his/her study notes on a given subject
WP8	A student wants to share his/her study notes on a given subject
WP9	A student wants to edit his/her study notes on a given subject
WP10	A student wants to delete his/her shared notes
WP11	A student wants to create a studying group
WP12	A student wants to invite people to his/her studying group
WP13	A student wants to chat with his/her studying group
WP14	A student wants to delete his/her studying group
WP15	A student wants to publish an announcement on the forum
WP16	A student wants to edit an announcement on the forum
WP17	A student wants to delete an announcement on the forum
WP18	A student wants to comment an announcement on the forum
WP19	A student wants to delete a comment on the forum
WP20	A student wants to share an announcement found on the forum
WP21	A student wants to find documents relative to a topic
WP22	A student wants to buy an academic book
WP23	A student wants to offer private tutoring lessons
WP24	A student wants to purchase a private tutoring lessons
WP25	A student wants to customize his profile
WP26	A student wants to log in to their account
WP27	A student wants to log out of their account

Table 2: World Phenomena

1.2.2 Shared Phenomena

ID	Description
SP1	The student creates an account on the system
SP2	The student deletes the account on the system
SP3	The student uploads a deadline in the system
SP4	The student moves a deadline in the system
SP5	The student edits a deadline in the system
SP6	The student deletes a deadline in the system
SP7	The student uploads the study notes in the system
SP8	The student uses the share function of notes provided by the system
SP9	The student edits the study notes in the system
SP10	The student deletes the study notes from the system
SP11	The student creates a study group in the system
SP12	The student invites people to the study group using the system
SP13	The student chats with the study group through the system
SP14	The student deletes the study group from the system
SP15	The student publishes an announcement on the system forum
SP16	The student edits an announcement on the system forum
SP17	The student deletes an announcement from the system forum
SP18	The student comments on an announcement in the system forum
SP19	The student deletes a comment from the system forum
SP20	The student shares an announcement using the system
SP21	The student searches a topic in the system forum
SP22	The student purchases an academic book through the system
SP23	The student offers private tutoring lessons in the system
SP24	The student books a private tutoring lesson using the system
SP25	The student customizes the profile in the system
SP26	The student logs in to their account
SP27	A student logs out of their account

Table 3: Shared Phenomena

1.3 Revision History

- Version 1.0 - May 4, 2025 - Initial release of the document.

1.4 Document Structure

This document is structured into 6 sections, each containing subsections that provide detailed information about the system. The sections are as follows:

- **Introduction:** This section provides an overview of the purpose, scope, and goals of the system. It also includes a description of the world phenomena and shared phenomena that the system will address.
- **Overall Description:** This section describes the product perspective, including scenarios and a domain model. It also provides a high-level overview of the system's functionality and user characteristics.
- **Specific Requirements:** This section details the external interface requirements, functional requirements, performance requirements, and security requirements of the system.
- **Effort Spent:** This section provides a summary of the effort spent by each team member on the project, including time spent on various tasks.
- **References:** This section lists all the references used in the document, including general references, websites, and tools.

2 Overall Description

2.1 Product Perspective

2.1.1 Scenarios

Scenario 1: A user creates an account and manages his own deadlines

Elir, a first-year Computer Science student, creates an account on StudentHub using his university email. After verification, he adds three academic deadlines to the Calendar: a Programming assignment, a Mathematics exam, and a group project meeting. He marks the group project as "shareable" for future teammates. Elir also adds a private reminder to call home on the weekend. Later, when his Programming assignment deadline is extended, Elir easily updates it in the system.

Scenario 2: A user creates an announcement on the forum

Davide T., an Economics student, searches StudentHub's Forum for a specific Econometrics textbook. Not finding it, he creates an announcement requesting "Advanced Econometrics by Rosiglioni D. - 7th Edition" with his budget. Two days later, Salvo comments on his post offering to sell his copy. They arrange the exchange through direct messaging and meet at the university library. Trevisan then marks his announcement as "Resolved" with a thank-you note.

Scenario 3: A user creates a study group and shares notes

Pantarotto A., a Biology student, uploads his Cell Structure notes to StudentHub's "Study Resources" section with relevant tags. He then creates a "Biology Midterm Prep" study group, making it discoverable to other Biology students. Alessio invites five classmates, three of whom accept immediately. Through the group chat, he shares his notes and the team plans their study sessions for the upcoming midterm.

2.1.2 Domain Model

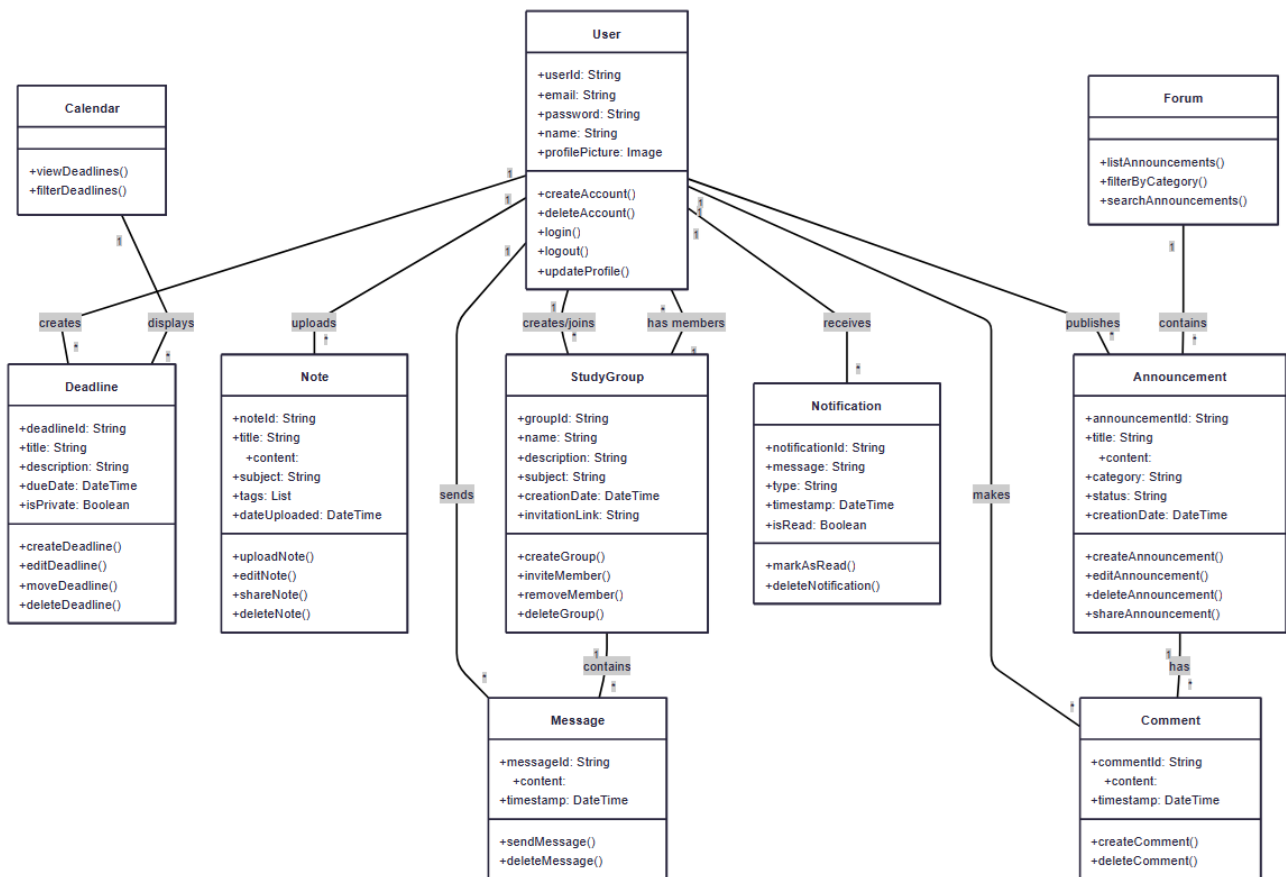


Figure 1: High level class diagram of the system

Domain Model Description:

This domain model represents the core structure of StudentHub, a platform designed to streamline student life management. The User entity is at the core of the system, with connections to all the major components. The Calendar interface allows users to create and manage deadlines, offering options for both private and shared academic tasks. The Note entity facilitates the uploading and sharing of study materials, organised by subject and tags.

StudyGroups facilitate collaboration between users, enabling members to exchange messages and coordinate study sessions. The creation of a group involves the sending of invitations to potential members. The Forum contains an Announcements section where users can post requests for books, offer tutoring services, or share event information. Should users wish to interact with announcements, they are able to do so through the 'Comments' function.

The model also includes a notification system that keeps users informed about deadlines, group invitations and forum activity. The relationships between entities reflect the platform's collaborative nature, demonstrating how users interact with content and each other.

This structure supports all of the identified goals of the RASD, including account management, deadline tracking, resource sharing, group collaboration, forum interactions, and marketplace functionality, creating a comprehensive ecosystem for student academic life management.

2.2 Product functions

This section describes the main functions of the system, focusing on the actions that users can perform and the interactions between them and the system.

2.2.1 Student Functions

- **Create an account:** a student provides the system his/her full name and valid institutional email, the student chooses a password and confirms it. The system sends a confirmation of entry email to validate. When confirmed, the account is created.
- **Delete an account:** the student enters his/her profile settings and clicks on the delete account button, the system displays a label where the student has to confirm the password. If the password is correct the account is deleted.
- **Log in:** the student enters his email and password. If the data are correct, he/she is logged in and accesses the global forum; otherwise, an error is displayed.
- **Log out:** The student clicks on the logout button on the screen, the system asks for confirmation, and if "yes" is clicked, the student is logged out.
- **Create a deadline:** the student clicks on the add deadline button on the calendar page, and the system displays a modal for the deadline name and description. The student enters the requested data and clicks "Publish".
- **Move a deadline:** the student clicks and drags with the mouse the deadline he/she wants to move and drops it on the proper date.
- **Publish some notes/an announcement:** the student on the proper page clicks on the new button and inserts his data, the student uploads eventual attachments. The student then clicks on publish.
- **Edit a deadline/notes/an announcement:** the student selects the item he/she wants to edit and clicks on the edit button. The student edits what he/she needs and confirms the new data by clicking on "Publish".
- **Delete a deadline/notes/an announcement:** the student selects the item he/she wants to delete and clicks on the delete button. A confirmation modal appears; the student clicks on confirm.
- **Send chat message on a study group:** The student goes to the study group page, types the message in the message box and then clicks send.
- **Add a comment to an announcement:** The student enters the comment in the comment box below the announcement and then clicks send.
- **Get the share link of an announcement:** The student clicks on the share icon on the announcement he/she wants to share and the link is copied.
- **Search a topic on the forum:** The student enters the topic in the search bar and clicks the search icon. The list of documents related to the searched topic will be displayed.

2.3 User Characteristics

This section describes the general characteristics of the intended groups of users.

2.3.1 Student characteristics

The students who will use this platform are usually enrolled at a university, so they are most likely to be between 18 and 30 years old and certainly have basic digital skills. A device with a stable connection is required to access the platform, as it all works by communicating with others. The platform provides translation in the most common languages, so users are only expected to have basic reading skills in one of the supported languages and do not require advanced technical knowledge beyond standard web and app usage. The platform should use a user interface with a high-contrast palette to improve the experience of visually impaired users and be compatible with screen readers (for example, using alternative text for images), ensuring accessibility for users with various disabilities. In general, it should be accessible to all types of users. Students are expected to access the platform regularly, either daily or weekly, to manage deadlines, share resources, and interact with study groups.

2.4 Assumptions, dependencies and constraints

ID	Description
A1	The platform is accessible via web browser on any device
A2	Students have valid institutional email addresses to register on the platform
A3	Students should use their correct personal data and not perform any kind of fraudulent or deceptive activity within the platform
A4	Each student has access only to the study groups he/she is a part of
A5	Students are enrolled at recognized Universities
A6	An institutional email allows the creation of at most one account
A7	Students are responsible of all the data they submit to the platform
A8	The system validates inputs (e.g. password criteria, and file size limits) before processing actions.
A9	The system maintains data integrity and consistency across all operations (e.g. when deleting an STU account, associated data are also removed).
A10	The platform relies on users having a stable internet connection during use
A11	Students are expected to keep their login credentials secure and confidential
A12	Forum posts, announcements, and notes shared by users should respect copyright and legal regulations
A13	The platform will only process and store data compliant with data protection regulations (e.g., GDPR)
A14	Notifications depend on users enabling them and providing correct contact information
A15	The system time zone defaults to the student's selected or device time zone for deadlines and schedules
A16	Translation services are provided by an integrated external API (e.g., Google Translate API, DeepL API)

Table 4: Domain assumptions

3 Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

The StudentHub platform will provide the following user interfaces:

- **Landing Page:** Displays general information about the platform and provides access to login and registration.
- **Dashboard:** Displays a summary of the user's calendar, deadlines, recent notes, and notifications.
- **Calendar Interface:** Shows shared and private deadlines in a calendar view with daily, weekly, and monthly views.
- **Study Notes Repository:** Interface for uploading, searching, and managing study notes.
- **Study Group Interface:** For creating, managing, and participating in study groups.
- **Forum Interface:** For creating, viewing, and interacting with announcements.
- **Marketplace Interface:** For listing and searching books and tutoring services.
- **User Profile Interface:** For managing personal information and preferences.

All interfaces must be responsive and accessible on desktop and mobile devices.

3.1.2 Hardware Interfaces

The system does not require specific hardware interfaces beyond standard computing devices with internet connectivity:

- Desktop computers
- Laptop computers
- Tablets
- Smartphones

3.1.3 Software Interfaces

The system will interface with the following software systems:

- **Email Service API:** For user verification and notifications
- **Database Management System:** For storing user data, notes, calendar events, and forum content
- **File Storage System:** For storing uploaded study materials
- **Authentication System:** For user login and security

3.1.4 Communication Interfaces

The system will utilize the following communication protocols:

- **HTTPS:** For secure data transmission between client and server
- **WebSocket:** For real-time notifications and messaging within study groups
- **SMTP:** For sending emails to users

3.2 Functional Requirements

3.2.1 User Management

ID	Description
FR1.1	The system shall allow users to create an account using their university email address
FR1.2	The system shall verify user email addresses through a confirmation link
FR1.3	The system shall allow users to log in to their account using their credentials
FR1.4	The system shall allow users to log out from their account
FR1.5	The system shall allow users to delete their account
FR1.6	The system shall allow users to reset their password through email verification
FR1.7	The system shall allow users to customize their profile information including: name, profile picture, field of study, and brief bio

Table 5: User Management Requirements

3.2.2 Calendar Management

ID	Description
FR2.1	The system shall provide a calendar interface with daily, weekly, and monthly views
FR2.2	The system shall allow users to create new deadlines with title, description, date, time, and category
FR2.3	The system shall allow users to categorize deadlines (e.g., exam, assignment, project)
FR2.4	The system shall allow users to set deadlines as private (visible only to themselves) or shared (visible to specified users or groups)
FR2.5	The system shall allow users to edit deadlines they have created, including changing all attributes
FR2.6	The system shall allow users to move deadlines to different dates/times through drag-and-drop or date selection
FR2.7	The system shall allow users to delete deadlines they have created
FR2.8	The system shall allow users to filter deadlines by category, date range, or visibility (private/shared)

Table 6: Calendar Management Requirements

3.2.3 Study Notes Management

ID	Description
FR3.1	The system shall allow users to upload study notes in various formats (PDF, DOC, TXT) with a maximum file size of 20MB
FR3.2	The system shall allow users to categorize notes by subject, course, and academic term
FR3.3	The system shall allow users to add tags to their notes for better searchability
FR3.4	The system shall allow users to share notes with specific users, study groups, or make them publicly available
FR3.5	The system shall allow users to edit metadata of their uploaded notes (title, description, tags)
FR3.6	The system shall allow users to update their notes by uploading a new version
FR3.7	The system shall allow users to delete their uploaded notes
FR3.8	The system shall provide a search function for finding notes by subject, tag, or content

Table 7: Study Notes Management Requirements

3.2.4 Study Group Management

ID	Description
FR4.1	The system shall allow users to create study groups with a name, description, and subject area
FR4.2	The system shall allow users to set study groups as public (discoverable by all users) or private (accessible by invitation only)
FR4.3	The system shall allow users to invite other users to join their study groups
FR4.4	The system shall allow users to accept or decline study group invitations
FR4.5	The system shall provide a messaging system within study groups
FR4.6	The system shall allow users to share notes and resources directly within study groups
FR4.7	The system shall allow study group creators to manage group members (add/remove)
FR4.8	The system shall allow study group creators to delete their groups
FR4.9	The system shall allow users to leave study groups they have joined

Table 8: Study Group Management Requirements

3.2.5 Forum Management

ID	Description
FR5.1	The system shall provide a forum where users can create announcements with title, content, and category
FR5.2	The system shall allow users to categorize announcements (e.g., book exchange, tutoring, events)
FR5.3	The system shall allow users to edit their own announcements
FR5.4	The system shall allow users to delete their own announcements
FR5.5	The system shall allow users to comment on announcements
FR5.6	The system shall allow users to edit and delete their own comments
FR5.7	The system shall allow users to mark announcements as resolved when applicable
FR5.8	The system shall allow users to share announcements via email or social media
FR5.9	The system shall provide a search function for finding announcements by category, keywords, or date

Table 9: Forum Management Requirements

3.2.6 Notification System

ID	Description
FR7.1	The system shall notify users of upcoming deadlines 24 hours before the deadline
FR7.2	The system shall notify users of new study group invitations
FR7.3	The system shall notify users of new messages in their study groups
FR7.4	The system shall notify users of comments on their forum announcements
FR7.5	The system shall notify users of responses to their marketplace listings
FR7.6	The system shall allow users to customize notification preferences (email, in-app, or both)
FR7.7	The system shall allow users to disable specific types of notifications

Table 10: Notification Requirements

3.3 Performance Requirements

3.3.1 Response Time

ID	Description
PR1.1	The system shall load web pages within 3 seconds under normal load conditions
PR1.2	The system shall process form submissions within 2 seconds
PR1.3	The system shall display search results within 3 seconds
PR1.4	The system shall upload files of up to 20MB within 10 seconds with standard internet connection
PR1.5	The system shall deliver notifications in real-time (within 5 seconds of the triggering event)

Table 11: Response Time Requirements

3.3.2 Capacity

ID	Description
PR2.1	The system shall support at least 1000 concurrent users
PR2.2	The system shall support up to 100,000 registered users
PR2.3	The system shall support up to 10,000 study groups
PR2.4	The system shall support up to 500,000 documents in the notes repository
PR2.5	The system shall support up to 100,000 forum announcements

Table 12: Capacity Requirements

3.3.3 Reliability

ID	Description
PR3.1	The system shall maintain 99.5% uptime in a year
PR3.2	The system shall recover from failures within 5 minutes
PR3.3	The system database shall be backed up daily with a retention period of 30 days
PR3.4	The system shall maintain data integrity across all operations
PR3.5	The system shall implement appropriate error handling to prevent crashes

Table 13: Reliability Requirements

3.3.4 Scalability

ID	Description
PR4.1	The system shall be designed to scale horizontally to accommodate increased user load
PR4.2	The system shall maintain performance levels when scaling up to 200% of the initial capacity
PR4.3	The system shall implement load balancing to distribute traffic efficiently
PR4.4	The system shall implement database sharding to manage data growth

Table 14: Scalability Requirements

3.3.5 Security

ID	Description
PR5.1	The system shall encrypt all user passwords using industry-standard hashing algorithms
PR5.2	The system shall enforce secure password policies (minimum 8 characters, including uppercase, lowercase, numbers, and special characters)
PR5.3	The system shall implement HTTPS for all data transmissions
PR5.4	The system shall implement protection against common web vulnerabilities (XSS, CSRF, SQL injection)
PR5.5	The system shall log all login attempts and security-related events
PR5.6	The system shall automatically log out inactive users after 30 minutes
PR5.7	The system shall implement rate limiting to prevent brute force attacks

Table 15: Security Requirements

4 Effort spent

4.1 Argentini Elir

Task	Effort spent (hours)
Introduction	3
Overall Description	1:30
Specific Requirements	2
Reasoning	3
Writing and Fixing	3
Graphic Adjustments	0
References	0
Total	12:30

4.2 Battaglia Salvatore

Task	Effort spent (hours)
Introduction	2
Overall Description	1:30
Specific Requirements	2
Reasoning	3
Writing and Fixing	1
Graphic Adjustments	2
References	1
Total	12:30

4.3 Pantarotto Alessio

Task	Effort spent (hours)
Introduction	0
Overall Description	0
Specific Requirements	2
Reasoning	3
Writing and Fixing	3
Graphic Adjustments	1
References	0
Total	9

4.4 Rosiglioni Diego

Task	Effort spent (hours)
Introduction	0
Overall Description	0
Specific Requirements	1
Reasoning	2
Writing and Fixing	1
Graphic Adjustments	4
References	0
Total	8

5 References

5.1 General References

- **Assignment instructions:** Teacher's shared project assignment.
"Esercitazione di Laboratorio RASD"
- **IEEE. (2023).** IEEE Recommended Practice for Software Requirements Specifications (IEEE 830-1998). Institute of Electrical and Electronics Engineers.
- **Latex documentation.** (2025). The $\text{\LaTeX}$ Project.
Retrieved from <https://www.latex-project.org/help/documentation/>
- **Example document:** Teacher Lukash Matteo's "Students & Companies - Requirements Analysis and Specification Document" document.
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5.2 Websites

- <https://www.latex-project.org/help/documentation/>
- <https://stackoverflow.com/questions>
- https://github.com/salvobat/RASD_GPOI
- <https://www.overleaf.com/learn>
- <https://www.mermaidchart.com/>
- <https://chatgpt.com/>

5.3 Tools

- **Visual Studio Code** - Code editor used for writing the document in LaTeX.
- **Overleaf** - Online LaTeX editor used for collaborative writing and editing of the document.
- **Mermaidchart** - Online tool used for creating the domain model diagram.
- **LaTeX Workshop** - VSCode extension used for LaTeX editing and compilation.
- **GitHub** - Version control system used for collaborative writing and editing of the document.
- **Live Share** - VSCode extension used for real-time collaboration on the document.
- **ChatGPT** - AI language model used for generating text and here used for a better LaTeX formatting.